

INTRODUCTION TO PROBABILITY THEORY

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INTRODUCTION TO PROBABILITY

What is probability? Probability is a measure of chance, how likely an event is to occur. I once had a friend who claimed that everything had a 50/50 chance of happening, it either happens or it doesn't. While amusing, this isn't quite how probability works. In a world filled with uncertainty, some events are clearly more likely than others, and probability gives us a way to quantify that likelihood.

Probability theory provides the mathematical framework for understanding and reasoning about randomness and uncertainty that characterise these events that happen in life. By assigning numerical values to uncertain events, we can analyse data, make predictions, and assess risk in a systematic way. This is especially important in finance, where asset prices fluctuate, interest rates change, and economic events unfold unpredictably. Probability allows us to model these uncertainties, estimate the likelihood of different outcomes, and make more informed decisions in the presence of risk.

BASIC DEFINITIONS AND SET OPERATIONS

To grasp the basics of probability theory, we must first familiarise ourselves with some definitions and set operations. This will grant us a solid understanding to approach probability problems.

The **outcome** of an experiment or trial is denoted by ω . The set of all possible outcomes of an experiment, referred to as the **sample space/probability space** is denoted by Ω . For example, a six-sided die has the sample space $\{1, 2, 3, 4, 5, 6\}$. Since probability quantifies the chance of an event occurring, we express it as a number between 0 and 1, where 0 denotes the event cannot happen, and 1 means it is certain to happen. The probability of an outcome is $P(\omega) \geq 0$, $\forall \omega \in \Omega$, and thus the sum of all possible outcomes add to 1, i.e., $\sum_{\omega \in \Omega} P(\omega) = 1$.

An **event** is a set of outcomes and a subset of the sample space. Continuing with the above example, the event of rolling an odd number on a die corresponds to the outcomes $\{1, 3, 5\}$. Therefore, the probability of an event A is denoted by $P(A) = \sum_{\omega \in A} P(\omega)$. If we have two events, A and B , then we denote the union as $A \cup B$, which

is the set of outcomes in event A , or in event B , or both. We denote the intersection as $A \cap B$, which is the set of outcomes in event A and B . The complement of A , denoted A^c or $\Omega \setminus A$, is the event "not A ". For events to be **mutually exclusive**, we have $A \cap B = \emptyset$, where $\emptyset$ is the empty set. For any mutually exclusive

events, $P\left(\bigcup_{i=1}^n A_i\right) = \sum_{i=1}^n P(A_i)$, the probability of at least one event A_i happening is the same as the sum of all probabilities of each event. Finally, a **random variable** is a function that maps each outcome, ω , in the sample space, Ω , into the set of real numbers. For example, in Article No.4, we utilised the random variable called the indicator variable, a binary dummy variable for an event A , which is defined as:

$$\mathbf{1}_A(x) = \begin{cases} 1 & \text{if } x \in A \\ 0 & \text{if } x \notin A \end{cases}$$

i.e., $\mathbf{1}_A = 1$ when A occurs and $\mathbf{1}_A = 0$ if A^c occurs. The neat trick here is that the expected value of $\mathbf{1}_A$ is $\mathbb{E}[\mathbf{1}_A] = P(A)$.

TYPES OF PROBABILITY

Understanding the distinct types of probability is essential for applying probability theory effectively.

1. Theoretical probability – based on the assumption that all outcomes are equally likely, for example, flipping a fair coin has a theoretical probability of landing on heads as 0.5 and landing on tails as 0.5 as well.
2. Experimental probability – derived from real life observations or actual experiments. Continuing with the coin example, if you flip a coin 100 times and it lands on tails 48 times, the experimental probability of landing on tails is $P(\text{tails}) = 0.48$.
3. Subjective probability – based on personal judgement or experience rather than calculations, reflecting an individual's belief about the likelihood of an event occurring.

AN EXAMPLE

I have always found that concepts are best understood through examples. For example, as someone who enjoys card games, let's consider a simple card game. There are 52 cards in a classic deck, each with 4 cards for each value, 2, 3, 4, 5, 6, 7, 8, 9, 10, $J.Q.K.A$, where J, Q, K, A represent the Jack, Queen, King, and Ace respectively. For

each selection, the cards are thoroughly shuffled, so that each card has a theoretical probability. You pick a card from the deck, and your opponent picks another without replacement. If you have the larger number, you win; if the numbers are equal, no one wins; and if the number is smaller, your opponent wins. What is the probability of winning?

Solution: We can approach this problem in a few different ways. One way is to consider all 13 different outcomes of your card. The card can have a value from 2 to 14 (from an Ace), with an equal probability of $\frac{1}{13}$. With a value of 2, the probability of winning is thus $\frac{0}{51}$, i.e., no chance. With a value of 3, the probability of winning is $\frac{1}{51}$, i.e., when your opponent picks a 2. $\dots$. With a value of 13, the probability of winning is $\frac{12}{51}$, i.e., when your opponent picks a 2, 3, $\dots$, 12. Thus, the probability of winning is $\frac{1}{13} \times (\frac{0}{51} + \frac{1}{51} + \dots + \frac{12}{51}) = \frac{1}{13} \times \frac{12 \times 13}{2 \times 51} = \frac{8}{17}$. Note that the summation from 0 to 12 can be easily found from the formula $\sum_{i=1}^n i = \frac{n}{2}(n+1)$.

Another way to approach the problem is to consider the 3 possible outcomes: A_1 your card is larger and hence you win; A_2 your card is the same and hence a draw; A_3 your card is smaller and hence you lose. By a symmetrical argument, we can clearly see that $P(A_1) = P(A_3)$, so we only need to figure out the probability of equal value, a draw, $P(A_2)$. After the random selection of your card, your opponent has only 3 cards they can select to result in a draw, among the remaining 51 cards. Thus, the probability of winning is $P(A_1) = \frac{(1-P(A_2))}{2} = \frac{(1-\frac{3}{51})}{2} = \frac{8}{17}$, a little simpler when calculating.

Probability questions are a staple of interviews, where they are often used to test your intuition and understanding of uncertainty.

CONCLUSION

Probability theory has applications in countless fields, from finance and engineering to medicine and artificial intelligence. As interest in AI (Artificial Intelligence) and machine learning continues to grow, the importance of probability has become increasingly apparent. Many machine learning algorithms rely on probabilistic methods to learn from data and make predictions, while concepts such as Bayesian inference are built directly upon the foundations of probability theory.

An introduction to probability theory helps us quantify uncertainty. By mastering its fundamental concepts, rules, and applications, we can enhance our ability to analyse data, assess risk, and make informed decisions.

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